| MCS-105 | CLOUD COMPUTING | L T P<br>3 1 0 | 4 Credits |
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**Course Objectives:**

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| **CO1** | Basics of cloud computing. |
| **CO2** | Key concepts of virtualization. |
| **CO3** | Different Cloud Computing services |
| **CO4** | Cloud Implementation, Programming and Mobile cloud computing |
| **CO5** | Cloud Backup and solutions |

## MODULE 1

Cloud Computing – History of Cloud Computing – Cloud Architecture – Cloud Storage Advantages of Cloud Computing, Disadvantages of Cloud Computing, Companies in the Cloud Today, Cloud Services, Pros and Cons of Cloud Service Development, Types of Cloud Service Development – Software as a Service – Platform as a Service – Web Services – On-Demand Computing – Discovering Cloud Services Development Services and Tools – Amazon Ec2 – Google App Engine – IBM Clouds

## MODULE 2

Overview of Cloud Networks, Network Types, LAN, gateways and Router, IP Classes and subnets, CIDR Utilities, Connection Management, Security groups, and Amazon elastic block storage EBS, Ubuntu in the cloud, Utilities, File system, Exploring Online Scheduling Applications – Exploring Online Planning and Task Management – Collaborating on Event Management, Collaborating on Databases – Storing and Sharing Files

## MODULE 3

Programming, Control structure, events based Init daemon , Configuring Apache , Directive, virtual hosts, MySQL server in cloud database, Backup and Recovery, database shading, EC2 Application, Web application design Focus on Search Engine, security, Firewall, Amazon Cloud.

**Text Books:**

- Cloud Computing: Principles and Paradigms, Editors: Raj KumarBuyya, James Bromberg, Andrej M Goscinski, Wiley, 2011.
- Visible Ops private Cloud: FromVirtulization to private Cloud in 4 Practical's steps,Andi Mann, Kurt Milne, Jeanne Mcrain IT Ptocess Institute , In: first edition( April8,2011)

**Reference Book:**

- Cloud Computing Explained: Implementation Handbook for Enterprises, John Rotan, Recursive Press (November 2, 2009)

**Course Objectives:**

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|---|---|
| **CO1** | Define Cloud Computing and memorize the different Cloud service and deployment models |
| **CO2** | To Implement Different Cloud Computing services |
| **CO3** | Use and examine different cloud computing services |
| **CO4** | Analyze the components of open stack & Google Cloud platform and understand Mobile Cloud Computing |
| **CO5** | Design & develop backup strategies for cloud data based on features |